SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA1612
COURSE CODE: Data Warehousing and Data Mining

Q1. Compare K-Means and Hierarchical Clustering

K-Means and Hierarchical Clustering are widely used unsupervised learning techniques. K-Means divides data into a predefined number of clusters by minimizing the distance between data points and their cluster centroids. Hierarchical clustering creates a tree-like structure of clusters called a dendrogram and does not require the number of clusters to be fixed initially.

Criteria	K-Means	Hierarchical Clustering
Basic idea	Partitions data into K clusters.	Builds nested clusters using a dendrogram.
Number of clusters	K must generally be specified.	Can be selected by cutting the dendrogram.
Speed	Fast for large datasets.	Usually slower for large datasets.
Cluster shape	Works best for compact, roughly spherical clusters.	Can identify different structures depending on linkage.
Real-world use	Customer segmentation, image compression.	Gene analysis, document grouping, taxonomy.

Q2. Compare K-Means and DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) groups points based on density. Unlike K-Means, it can identify noise and does not require the number of clusters in advance. Therefore, DBSCAN is useful when clusters have irregular shapes or when outliers are important.

Criteria	K-Means	DBSCAN
Cluster formation	Centroid-based.	Density-based.
K requirement	Requires K.	Does not require K.
Outliers	Sensitive to outliers.	Explicitly identifies noise/outliers.
Cluster shape	Best for spherical clusters.	Can detect arbitrary-shaped clusters.
Example	Market/customer segmentation.	Geographical hotspot and anomaly detection.

Q3. Compare Hierarchical Clustering and DBSCAN

Criteria	Hierarchical Clustering	DBSCAN
Approach	Creates a hierarchy of clusters.	Forms clusters from dense regions.
Parameters	Choice of linkage and distance	Mainly eps and minimum

	metric.	points.
Outliers	May be affected by noisy data.	Handles noise explicitly.
Visualization	Dendrogram provides a clear hierarchy.	Usually visualized using scatter plots.
Best suited for	Small/medium datasets and hierarchical relationships.	Spatial data and irregular clusters.

Q4. Real-World Applicability of Clustering Algorithms

Customer Segmentation: K-Means groups customers based on spending, frequency and behavior, helping businesses create targeted offers.

Fraud and Anomaly Detection: DBSCAN can separate dense normal behavior from isolated observations that may represent unusual transactions.

Image Processing: K-Means can cluster similar pixel colors and is commonly used for image compression and segmentation.

Healthcare and Bioinformatics: Hierarchical clustering can group patients or biological samples according to similarity and reveal relationships.

Geographical Analysis: DBSCAN can identify dense locations such as accident hotspots, delivery demand zones or areas of high activity.

Q5. Critical Analysis and Conclusion

No single clustering algorithm is best for every dataset. K-Means is computationally efficient and easy to interpret, making it suitable for large datasets with relatively compact clusters. Hierarchical clustering is valuable when relationships between groups need to be explored at multiple levels. DBSCAN is preferable when the dataset contains noise, outliers, or irregularly shaped clusters.

For a practical data-analysis project, the algorithm should be selected after examining the data distribution, scale, expected cluster structure and presence of outliers. Visualizations such as scatter plots, cluster maps, and dendrograms make the results easier to interpret. Evaluation measures such as silhouette score can also help compare clustering quality.

Overall, clustering algorithms are highly applicable to real-world unsupervised learning problems. A comparative approach improves decision-making because it considers both algorithmic strengths and limitations rather than selecting a method without examining the characteristics of the data.

Summary Comparison

Algorithm	Type	Needs K?	Handles Noise?	Typical Applications
K-Means	Centroid-based	Yes	No	Customer segmentation, image compression

Hierarchical	Hierarchy-based	No (initially)	Limited	Gene analysis, document grouping
DBSCAN	Density-based	No	Yes	Anomaly detection, spatial hotspots

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand